

2 (**short**) Figure 2 shows a light circular pulley of radius R mounted on a vertical mast by a frictionless pivot at point P. A mass M is suspended by a cable which is tethered to the mast at Q, a distance $2R$ below the pivot. At point X, halfway between P and Q, calculate the axial force, shear force and bending moment in the mast, making clear the direction in which they act. [10]

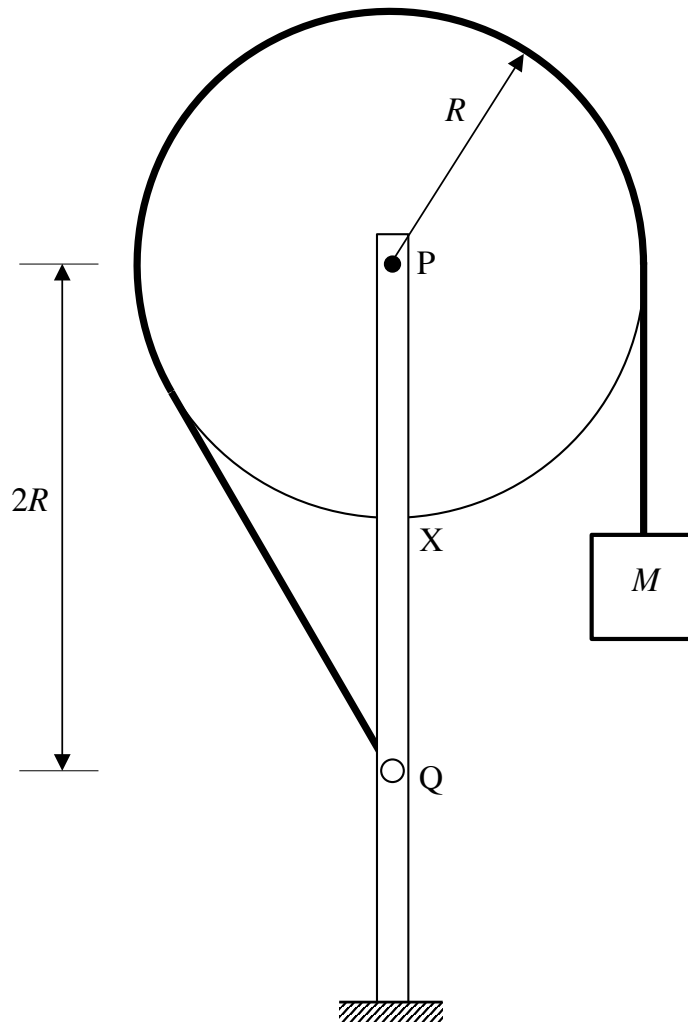


Fig. 2